Investigation of Gryllodes sigillatus Weight and ability to attrcatt females

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Introduction

sound dynamics within Gryllodes sigillatus Weight are extremely important and play a vitol role in enabling tem to mate with a female s there physiological structure influences the rate of which they are able to mate within a dangerous ecosystem in order to live on. this insect is sexually dimorphic and are prone to grow in tropical urban ecosystems around the world.

null hypothesis - To investigate if mass/sound play a massive role in Gryllodes sigillatus ability to sing and intern ability to mate.

Furthermore, there are relationships between song frequency patterns (whilst peak frequency being 6800 Hz pulse and a period of 150 ms) with male phenotype Gryllidae and female preference for male synthetic calling songs associated with different male body sizes; when looking at morphological traits related to calling patterns(Champagnon, J. and Cueva del Castillo, R et, l, 2008). This is measured via Aviso ft SAS Lab Pro software. Combining data analyse and biological investigation to see these differences is key to understanding if the null hypothesis stated plays an effect on the population f males that are able to mate and at the level that which they are able too. (Bateman, P.W. and MacFadyen, D.N.et . el, 1999)

```
plot_one
```

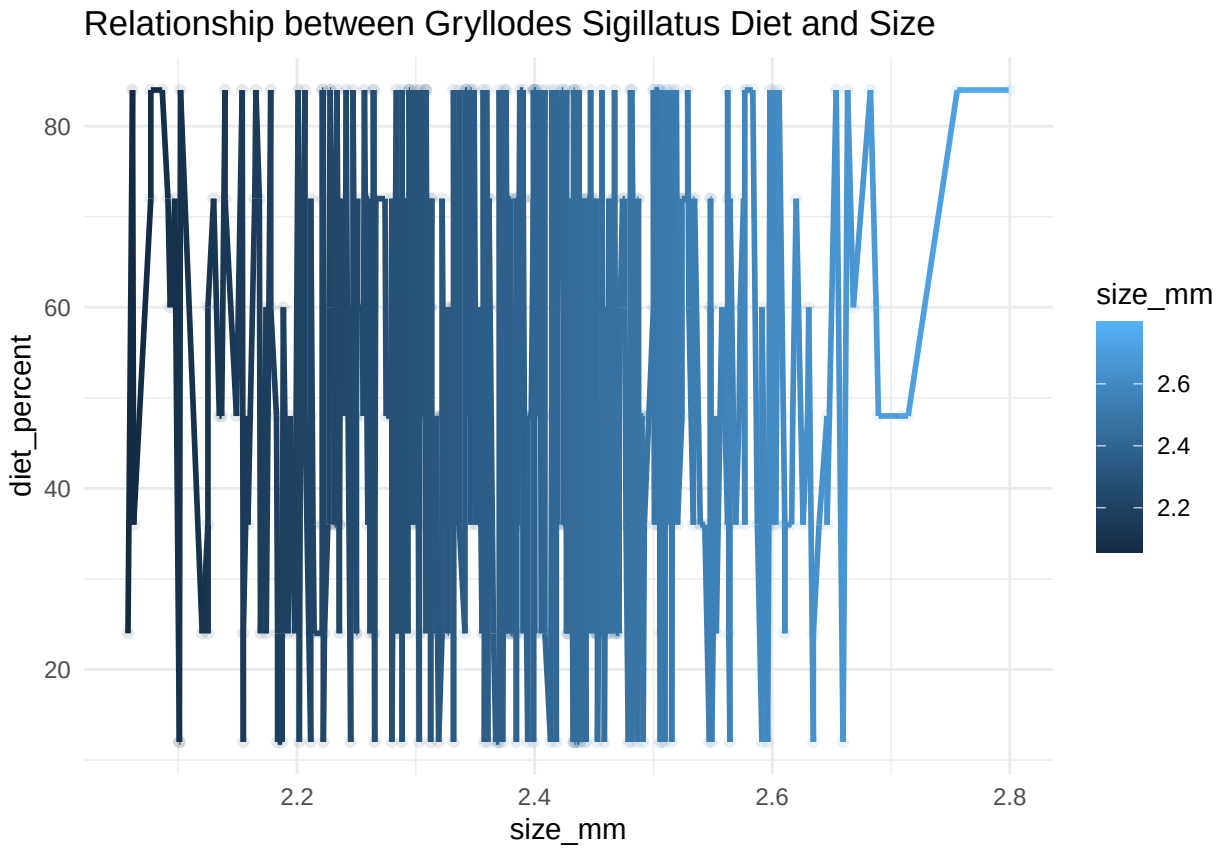


figure1 - line graph comparing the relationship of diet effecting size on a line graph with connecting lines to help track weight against diet.size is measured within a colour gradient to help visually see and express the relationship reported.

A line graph was used to display parameters briefly on a supportive metric scale that is confluent when being compared together. In my analyses, diet was not a strong enough supporting variable to truly support the null hypotheses as the statement of the graph doesn't convey smooth gradual lines. initially may not be the strongest representation. However, from this graph there ate some strong relationships regarding the percentage of food eaten relating to the size that a Gryllodes sigillatus will hold. with high volatility and high noises of frequency there is a lack of information to deduce from this and put towards my null hypothesis. poor correlation without any additional stats testing is (F-statistic: 0.06246 on 1 and 537 DF, p-value: 0.8027) not statistically significant at a conventional threshold. undergoing ANOVA testing produced (f-value 0.0625). taking these relus into significant account we can see that size increase with diet but cannot state if their ability to produce sound is ever so slightly effected by this.

Mann-Whitney U test

Null hypothesis the median difference between pairs of observation is 0 The alternative hypothesis – the median difference between pairs of observation is not 0

This test is used to compare 2 samples in represented of t-test where both sets of variables are normally distributed. The man Whitney test produces a statistical measurements for multiple variables within the test used to measure level of significance. I started of using the Shapiro t-test against diet and size. Here (W = 290521, p-value $<$ 2.2e-16) shows solidity and results gathered are reliable with an extremely convincing p value close enough to 0.

concluded: "the results of this data was poor and defective illustrating not only a weak visual representation but weak statistical values separately the tests used are strong indications that the data holds a positve value"

```
plot_two
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

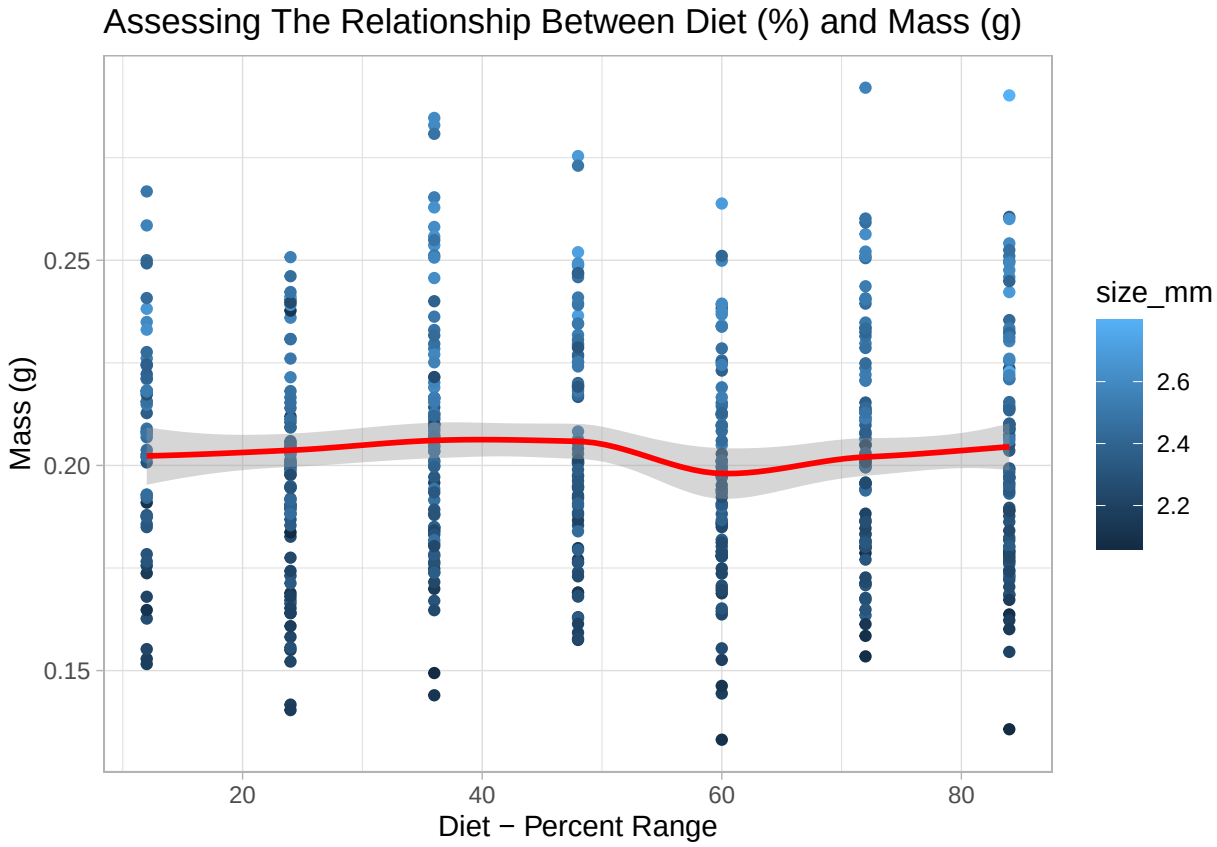
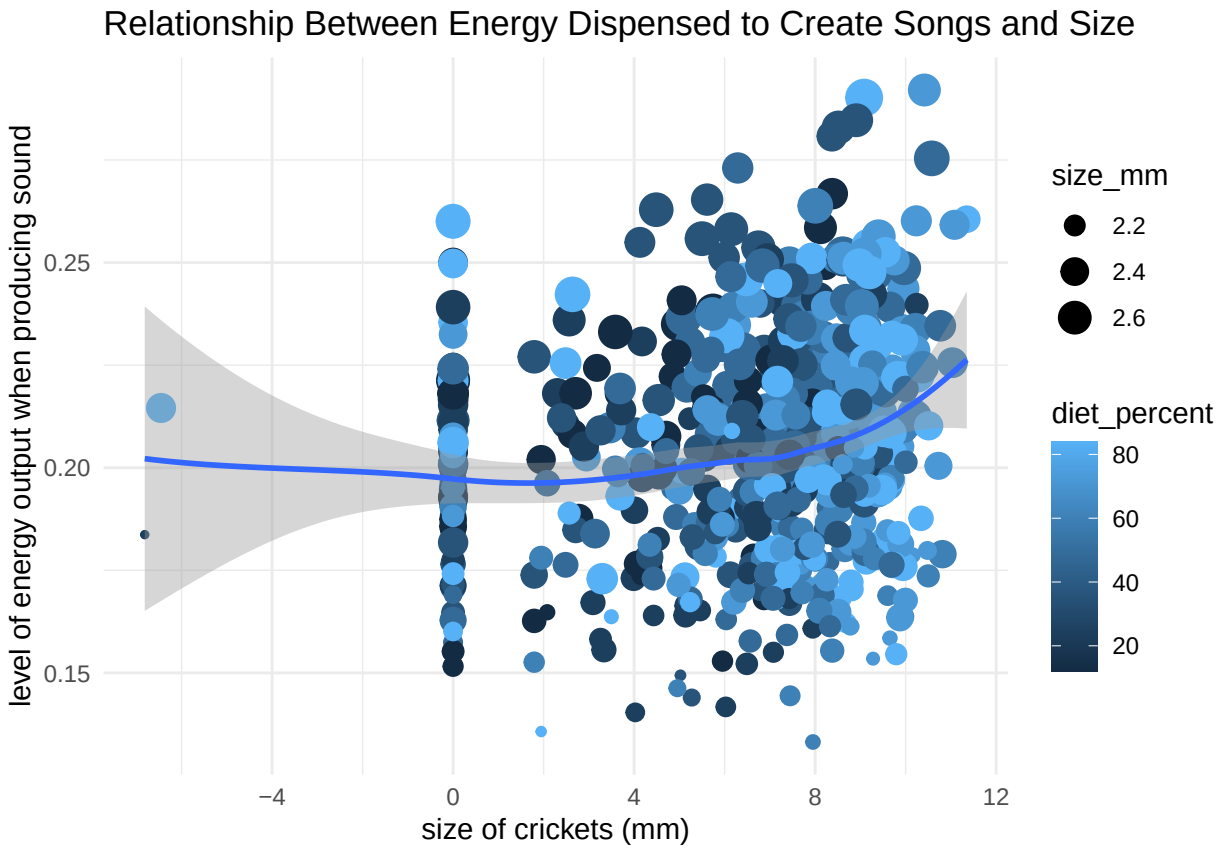


figure2 -scatter plot showing mass and diet in Gryllodes sigillatus crickets. vertical plots with multiple categorical variables with a line of standard error and colour gradient size. vertical of graph against both axis holds no correlation between each other.P-value:0.0625 showing weak statistical significance.

scatter plot was used as its a quick and effective method of showing communications between 2 variables. if successful the detection of outlines can indicate errors or anomalies when handling large data. the data visual used to support the null hypothesis was not effective but the graph used to try and exploit this data conveying a use of different categorical variables should be used to exploit a reliable statistical measurement. (Density at 95% confidence is -16%.Multiple R-squared: 1.845e-05) Adjusted R-squared: -0.001844 illustrating model performs worse than a model with no predictors and as a result the model is not a good fit for what i am trying to display.(F-statistic: 0.009906 on 1 and 537 DF) and a (p-value: 0.9208).

```
plot_three
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



figur3 - complex scatter graph distributing energy output and size. size is spread across holding different weight and sizes visibly to show direct differences between the quantity of Gryllodes sigillatus crickets. Size has almost a direct correlation to the amount of energy expended from Gryllodes sigillatus cricket. standard error f progression line produced to further examine reliability and its significannce in respects to how good of a fit it is.

A scatter plot with a size gradient and a line of best fit to convey lineage. the model summery conveyed that the coefficient was estimated to be (1.286546 whilst SE lied at 0.006957). there is a positive take hoeme on the coefficient as it conveys every unit in the x axis works conventionally with the y axis. as a result of this as size increases so does the ability to produce greater colume of music.intercepts was held at (2.3531797) in regards of a residual 2.5% confidence interval. (SE - 0.006957). This in tern conveys extrapolating data of Gryllodes sigillatus crickets hold meaning full data even if the weight is at 0.

subsequently, in our alternative hypothesis as the type of graph used to convey my data we holds a (p-value: 3.937e-07) illustrating that the observed data is unlikely to undergo the assumption of the null hypothesis. since the p value here is ($p<0.05$) lacks statistical significance.(ANOVA) F-value 26.376 with a 537 degree of freedom and a mean weight of (mean: 0.45167)

In conclusion - results displayed with the statistical measures that are must undergo further analyses and screening as it contests against other research that size does play a greater role as it allows more volume to be because (their body weight helps sustain sounds for longer) as singing is a high energy demanding action to perform in crickets.

Refferences

Champagnon, J. and Cueva del Castillo, R., 2008. Female mate choice, calling song and genetic variance in the cricket, Gryllodes sigillatus. Ethology, 114(3), pp.223-230. Bateman, P.W. and MacFadyen, D.N., 1999. Mate guarding in the cricket Gryllodes sigillatus: influence of multiple potential partners. Ethology, 105(11), pp.949-957.